Types of Chemical Reactions

There are four major types of chemical reactions that we will examine this year:

- Synthesis (or Addition)
- Decomposition
- Single Displacement
- Double Displacement.

We will also discuss *Combustion* reactions (which don't fit nicely into any of the above categories)

Synthesis

➢ Two or more reactants combine to form a single product.

➢ The general form is: $A + B \rightarrow AB$

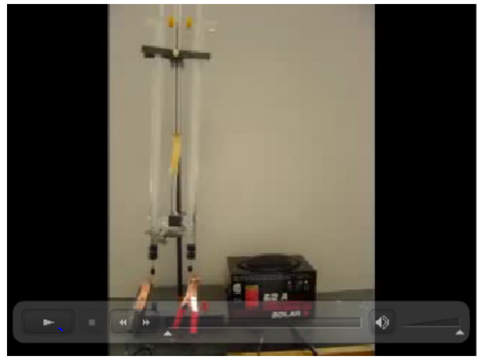
Ex. $2\,Mg_{(s)} + O_{2(g)} \xrightarrow{\Delta} 2\,MgO_{(s)}$

Ex. $2\,Na_{(s)} + Cl_{2(g)} \longrightarrow 2\,NaCl_{(s)}$

Decomposition

- A single compound is broken down into multiple products.
- This is the reverse of a synthesis reaction.
- The general form is:

$AB \rightarrow A + B$

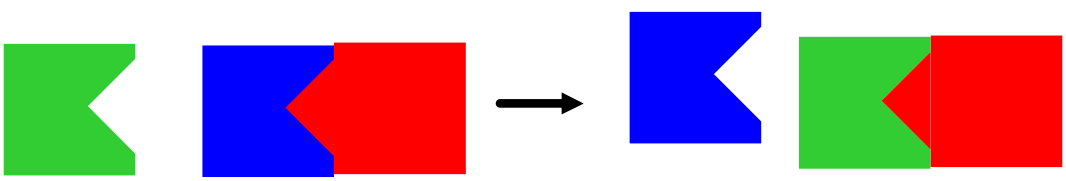


Ex. $2\ H_2O_{(l)} \xrightarrow{12\ volts} 2\ H_{2(g)} + O_{2(g)}$

http://www.youtube.com/watch?v=ZnojWOhg6Ww&safety_mode=true&persist_safety_mode=1

Single Displacement

- One element replaces another similar element in a compound.
- The general form is: $A + BC \rightarrow AC + B$

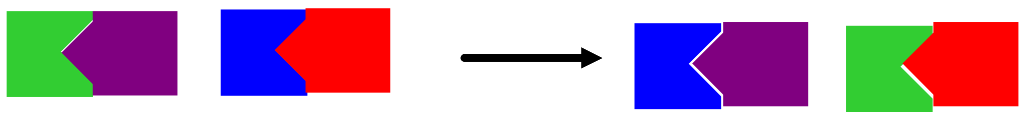


Ex. $Mg_{(s)} + 2\ HCl_{(aq)} \rightarrow MgCl_{2(aq)} + H_{2(g)}$

Ex. $\underline{2}\ NaI_{(aq)} + \underline{\quad}\ F_{2(g)} \rightarrow 2\ NaF_{(aq)} + I_2$

Double Displacement

- Two reactants recombine to form two new compounds.
- The general form is: AB + CD → AD + CB



Ex. $2\ KI_{(aq)} + Pb(NO_3)_{2(aq)} \rightarrow PbI_{2(s)} + 2\ KNO_{3(aq)}$

$$2\ Fe_{(s)} + O_{2(g)} \rightarrow 2\ FeO_{(s)}$$

$$2\ HNO_{3(aq)} + Cu_{(s)} \rightarrow Cu(NO_3)_{2(aq)} + H_{2(g)}$$

$$H_2SO_{4(aq)} + 2NaOH_{(aq)} \rightarrow Na_2SO_{4(aq)} + 2H_2O_{(l)}$$

$$Zn_{(s)} + 2HCl_{(aq)} \rightarrow ZnCl_{2(aq)} + H_{2(g)}$$